

Manual

Graphic Process Analysis (MOC)

PDV-GPA 8.1

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1 Grafische Prozessanalyse - Übersicht

Summary

Possible fields of application

This function package includes the graphic evaluations showing the measured values over time. The data lines can be selected separately and combined individually to be compared in diagrams. Binary process values (events) can also be presented in the diagrams. In addition, it is also possible to use the correlating MES data, such as order number, article number, and batch number as selection criteria and to show the machine statuses that are available at the machines at the same time.

Integration

A machine interface for data collection in the HYDRA Process Communication Controller (PCC) as well as the licenses for process data collection and processing (PDV-PDM and PDV-VRP) are required for using this function package.

Functions

Efficient functions to analyze process values in detail in correlation with other process values as well as logistical data from the production process

- Presentation (line chart) of the saved process values of a machine within a selectable period of time
- Presentation of several diagrams in one report for correlative considerations of process values of a machine
- Show/hide the machine status development in form of a time profile within the selected period
- Show/hide batch-related data in the selected period
- Show/hide selected process events within the selected period in form of a bit trace

2 Graphic Process Analysis (based on machines)

Summary

Menu	Quality Management → Process Analysis → Graphic Process Analysis (based on machines)
Transaction code	gpdam
Function authorization	gpda

The graphic process analysis is used for the process analysis in quality management.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other. In addition, the measured values can be displayed together with the machine status or binary process events. The selected process value lines and events can also be saved together as a compilation under a freely available name so that analyses that recur often can be called up easily.

Integration

The function displays collected measured values of the process characteristics, events and machine statuses. These values must have been previously collected in the system.

Selection criteria

The following selection criteria are available in the application:

Machine

The required workplace can be searched specifically in the "general" index tab using the pool of workplaces application.

Time domain from – until

Specifies a specific period of time for the evaluation.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Tag type + Tag ID

Selection of the tag type (in a user-friendly manner by a drop-down list) and the tag ID.

Consider long-term data

Long-term data may also be taken into account by clicking this checkbox.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Toolbar



Show process values

Show or hides the analog trace.



Show machine status

Shows or hides the machine status (machine time profile). This is shown in the lower section of the window.



Show events

Shows or hides the bit trace (events). This is shown in the lower section of the window.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

If the machine time profile is shown, the user also can carry out selections for the graphic of the machine time profile there. In the machine time profile, the machine status is always displayed. If resource statuses are used at the same time, they can also be shown.

If the events are shown, individual events can be selected there.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

3 Graphic Process Analysis (based on orders)

Summary

Menu	Quality management → Process analysis → Graphic process analysis (based on orders)
Transaction code	gpdao
Function authorization	gpdao

The graphic process analysis is used for the process analysis in quality management.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time period and selection criteria and to compare them with each other.

Integration

The function displays the selected and collected measured values of the process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

MES order number

Selects a specific operation (MES order number)

Batch

Selects a batch number

Time domain from – until

Selects a specific period of time

Tag type + Tag ID

Selection of the tag type (in a user-friendly manner by a drop-down list) and the tag ID.

Consider long-term data

Long-term data may also be taken into account by clicking this checkbox.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support.

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!

4 Graphic Process Analysis (based on batches)

Summary

Menu	Quality Management → Process Analysis → Graphic Process Analysis (based on batches)
Transaction code	gpdac
Function authorization	gpdac

The graphic process analysis is used for the process analysis in quality management.

Usage

The graphic process analysis provides a powerful graphic analysis tool for displaying process values read out of the database based on time periods and selection criteria and to compare them with each other.

Integration

The function displays the selected and recorded measured values of the process characteristics. These values must have been previously collected in the system.

Selection parameters

The following selection criteria are available in the application:

Batch number

Specifically selects a produced batch.

Alternative batch number 1

Selects the alternative batch number 1 at the produced batch

Period from – until

Selects a period of time

Tag type + Tag ID

Selects the tag type (in a user-friendly manner using a drop-down list) and the tag ID.

Consider long-term data

Long-term data may also be taken into account by clicking this checkbox.

Process parameters

The process parameters drop down list includes a selection of process parameters.

Please note that all data used for these evaluations/reports are kept in the server memory. Consequently, it is not recommendable to select too large periods and data sets. In addition, data might no longer be displayed clearly in graphics without additional functions, such as the zoom.

Technical background: every time data is requested in client applications, this data will be stored in the server memory and only then it will be transferred to the client. If you require more memory space, the memory reserved for the Java application needs to be increased accordingly on the server. Please refer to the technical documentation or contact MPDV Support

Detail applications

The graphics of the selected area are displayed in the graphic process analysis area. The buttons



can be used to insert another area for the graphic process analysis in which a new process parameter can be selected. Or an area inserted for the graphic process analysis is removed.

A special feature of this graphic is the ability to zoom in and out using keyboard and mouse combinations. To zoom in on the graphic, press the SHIFT key and hold it down and then use the left mouse button to highlight an area of the graphic. After the key is released, this area is enlarged in the display.

To zoom out again, press and hold the ALT key down. Now the mouse pointer appears as a magnifying glass with a minus sign. By clicking in the graphic, the view zooms out again and the image appears smaller.

Display configuration

The following configuration options can be set for each chart:

Checkbox “Legend”

A legend may be shown for the data displayed in the diagram by checking this option. In case a legend is shown, axis labeling is automatically hidden to save space.

Checkbox “Show scale”

The scale of the diagram can be shown or hidden by selecting this checkbox.

Checkbox “Show character designation”:

This checkbox specifies that the long characteristic names are shown for the legend and axis labeling instead of the short ID. Please note: labeling is not updated until data has been refreshed!